DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26	Year: Second Year	Term: I
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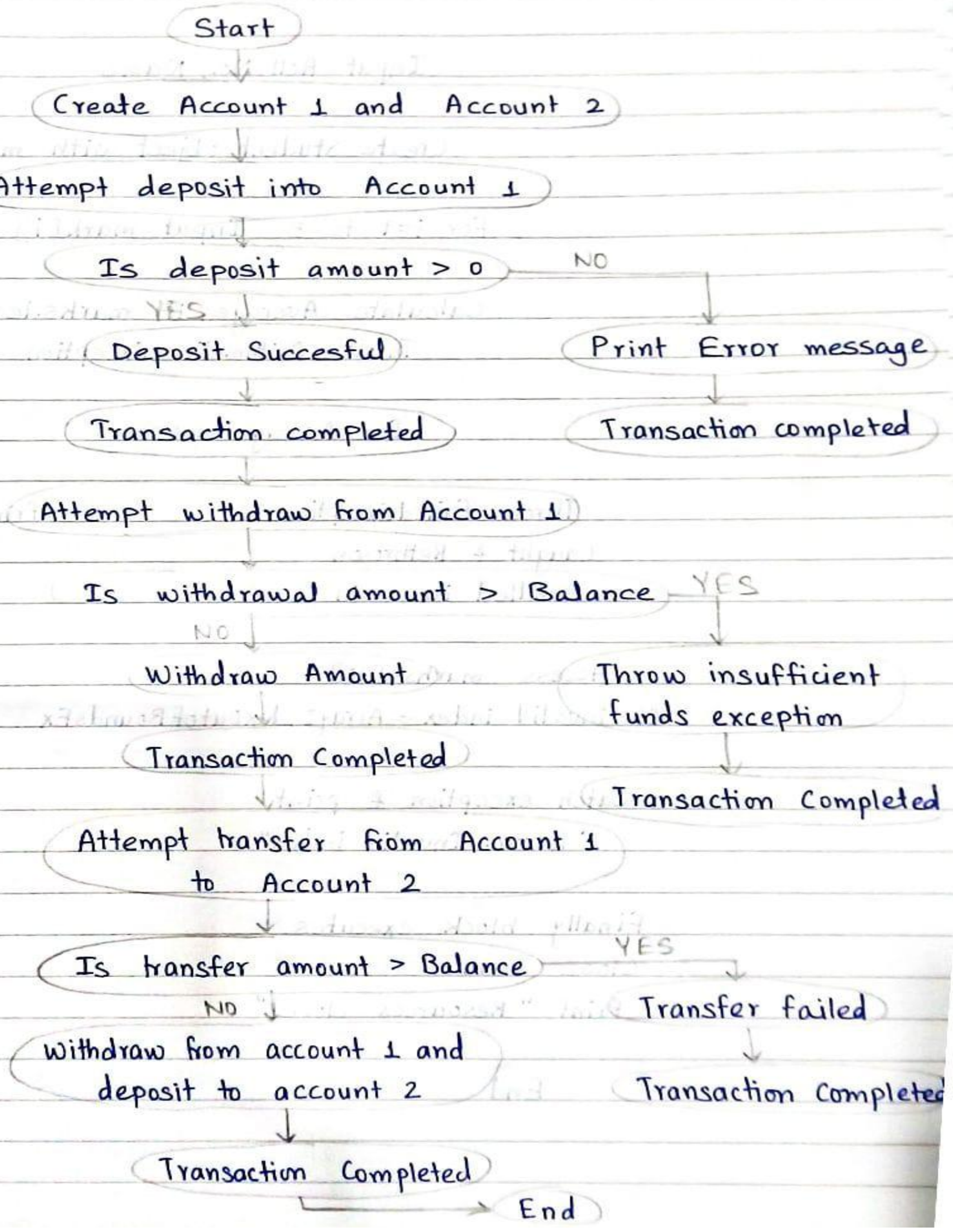
Subject: Java		
Assignment No.:	Title: Exception Handling	
Date: 25/08/2025		

Banking Application



```java
import java.util.Scanner;

class InsufficientFundsException extends Exception
{
    public InsufficientFundsException(String message)
    {
        super(message);
    }
}

class BankAccount
{
    private String accountNumber;
    private double balance;

    public BankAccount(String accountNumber, double balance)
    {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }

    public void deposit(double amount)
    {
        if (amount <= 0)
        {
            throw new IllegalArgumentException("Deposit amount must be greater than 0");
        }
        balance += amount;
        System.out.println("Deposited: " + amount + " | New Balance: " + balance);
    }

    public void withdraw(double amount) throws InsufficientFundsException
    {
        if (amount > balance)
        {
            throw new InsufficientFundsException("Not enough balance to withdraw " + amount);
        }
        balance -= amount;
        System.out.println("Withdrawn: " + amount + " | Remaining Balance: " + balance);
```

```java
    }

    public void transfer(BankAccount target, double amount)
    {
        try
        {
            this.withdraw(amount);
            target.deposit(amount);
            System.out.println("Transferred: " + amount + " from " + accountNumber + " to " + target.accountNumber);
        }
        catch (IllegalArgumentException e)
        {
            System.out.println("Transfer Failed: " + e.getMessage());
        }
        catch (InsufficientFundsException e)
        {
            System.out.println("Transfer Failed: " + e.getMessage());
        }
        catch (Exception e)
        {
            System.out.println("Unexpected Error: " + e.getMessage());
        }
        finally
        {
            System.out.println("Transaction Completed");
        }
    }
}

public class BankingApp
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account Number for Account 1: ");
        String accNum1 = sc.nextLine();
        System.out.print("Enter Initial Balance for Account 1: ");
        double bal1 = sc.nextDouble();
```

```java
        BankAccount acc1 = new BankAccount(accNum1, bal1);

        sc.nextLine();

        System.out.print("Enter Account Number for Account 2: ");
        String accNum2 = sc.nextLine();
        System.out.print("Enter Initial Balance for Account 2: ");
        double bal2 = sc.nextDouble();
        BankAccount acc2 = new BankAccount(accNum2, bal2);

        try
        {
            System.out.print("Enter amount to deposit in " + accNum1 + ": ");
            double depAmt = sc.nextDouble();
            acc1.deposit(depAmt);
        }
        catch (Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
        finally
        {
            System.out.println("Transaction Completed");
        }

        try
        {
            System.out.print("Enter amount to withdraw from " + accNum1 + ": ");
            double wAmt = sc.nextDouble();
            acc1.withdraw(wAmt);
        }
        catch (InsufficientFundsException e)
        {
            System.out.println("Error: " + e.getMessage());
        }
        finally
        {
            System.out.println("Transaction Completed");
        }
```

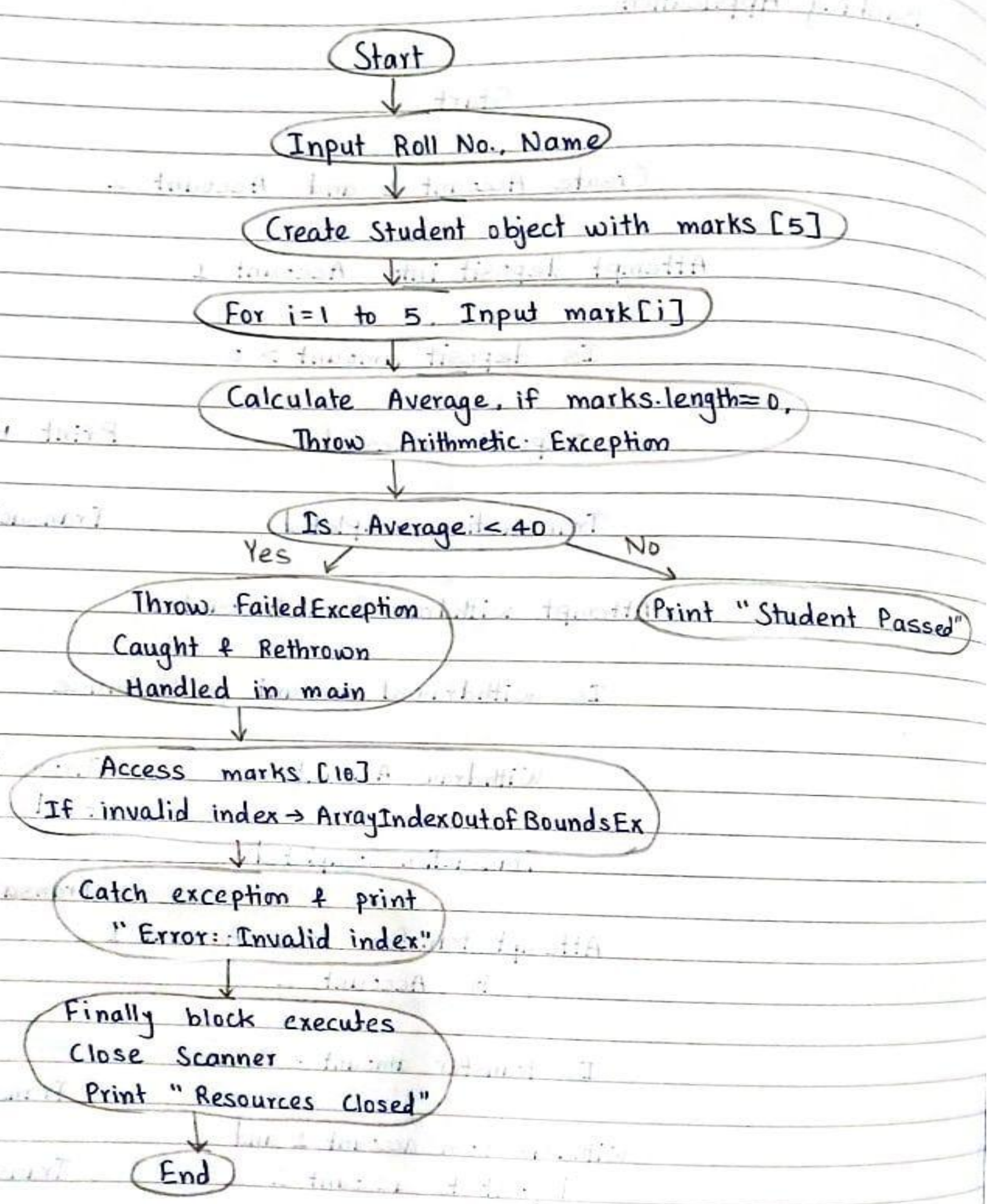
```java
            System.out.print("Enter amount to transfer from " + accNum1 + " to " + accNum2 + ": ");
            double tAmt = sc.nextDouble();
            acc1.transfer(acc2, tAmt);

        sc.close();
    }
}
```

Output:

```
Enter Account Number for Account 1: A
Enter Initial Balance for Account 1: 5000
Enter Account Number for Account 2: B
Enter Initial Balance for Account 2: 2000
Enter amount to deposit in A: 2000
Deposited: 2000.0 | New Balance: 7000.0
Transaction Completed
Enter amount to withdraw from A: 1000
Withdrawn: 1000.0 | Remaining Balance: 6000.0
Transaction Completed
Enter amount to transfer from A to B: 1500
Withdrawn: 1500.0 | Remaining Balance: 4500.0
Deposited: 1500.0 | New Balance: 3500.0
Transferred: 1500.0 from A to B
Transaction Completed
```

Student Result Application



```java
import java.util.Scanner;

class FailedException extends Exception
{
    public FailedException(String message)
    {
        super(message);
    }
}

class Student
{
    int rollNo;
    String name;
    int[] marks = new int[5];

    public Student(int rollNo, String name)
    {
        this.rollNo = rollNo;
        this.name = name;
    }

    public double calculateAverage()
    {
        int sum = 0;
        for (int m : marks)
        {
            sum += m;
        }
        if (marks.length == 0)
        {
            throw new ArithmeticException("Division by zero while calculating average");
        }
        return (double) sum / marks.length;
    }

    public void checkEligibility() throws FailedException
    {
        double avg = calculateAverage();
```

```java
        if (avg < 40)
        {
            throw new FailedException("Student " + name + " failed with average: " + avg);
        }
        else
        {
            System.out.println("Student " + name + " passed with average: " + avg);
        }
    }

    public int getMarks(int index)
    {
        return marks[index];
    }
}

public class StudentResultApp
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        try
        {
            System.out.print("Enter Roll No: ");
            int roll = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Name: ");
            String name = sc.nextLine();

            Student s = new Student(roll, name);

            try
            {
                for (int i = 0; i < 5; i++)
                {
                    System.out.print("Enter mark " + (i + 1) + ": ");
                    s.marks[i] = sc.nextInt();
                }
```

```java
                try
                {
                    s.checkEligibility();
                }
                catch (FailedException fe)
                {
                    System.out.println("Eligibility Check Failed: " + fe.getMessage());
                    throw fe;
                }

            }
            catch (FailedException fe)
            {
                System.out.println("Handled rethrown exception in main: " + fe.getMessage());
            }

            try
            {
                System.out.println("Accessing invalid mark: " + s.getMarks(10));
            }
            catch (ArrayIndexOutOfBoundsException e)
            {
                System.out.println("Error: Invalid index - " + e.getMessage());
            }

        }
        catch (Exception e)
        {
            System.out.println("Unexpected Error: " + e.getMessage());
        }
        finally
        {
            sc.close();
            System.out.println("Resources closed.");
        }
    }
}
```

Output:

```
Enter Roll No: 1
Enter Name: Riya
Enter mark 1: 70
Enter mark 2: 60
Enter mark 3: 75
Enter mark 4: 88
Enter mark 5: 80
Student Riya passed with average: 74.6
Error: Invalid index - Index 10 out of bounds for length 5
Resources closed.
```